BlockBuild Research

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1 Introduction

In the following paper, you will find a study of a neural network that predicts the EUR - USD exchange rate with the highest accuracy. For this purpose, three structures were selected, from which one is chosen to be tested and evaluated in terms of its accuracy. The training and testing processes were carried out using BlockBuild.

1.1 Neural Networks

1.1.1 Model 1

Structure: Conv1d(In: 11, Out: 3, Kernel Size: 3, Stride: 1, Padding: 1) → Conv1d(In: 3, Out: 1, Kernel Size: 3, Stride: 1, Padding: 1) → AdaptiveAvgPool1d(Output: 1) → Linear(In: 1, Out: 32) → ReLU → Dropout(Probability: 0.2) → Linear(In: 32, Out: 1); Fig. 1
The model uses one-dimensional convolutional layers to extract local dependencies between the input macroeconomic indicators, followed by pooling and fully connected layers for final regression.

1.1.2 Model 2 - Classical Multilayer Perceptron

Structure: Linear(In: 11, Out: 32) → ReLU → Linear(In: 32, Out: 64) → ReLU → Linear(In: 64, Out: 1); Fig. 1
A standard multilayer neural network that creates nonlinear relationships between the input economic data for the EUR–USD exchange rate. It is characterized by high stability and reduces the risk of mode collapse.

1.1.3 Model 3

Structure: LayerNorm(Normalized Shape: 11) → Linear (In: 11, Out: 16) → ReLU → Linear(In: 16, Out: 8) → ReLU → Linear(In: 8, Out: 1); Fig. 1
The model combines input data normalization (which is important due to their different permissible ranges) with a nonlinear network, improving training stability and reducing the risk of mode collapse.

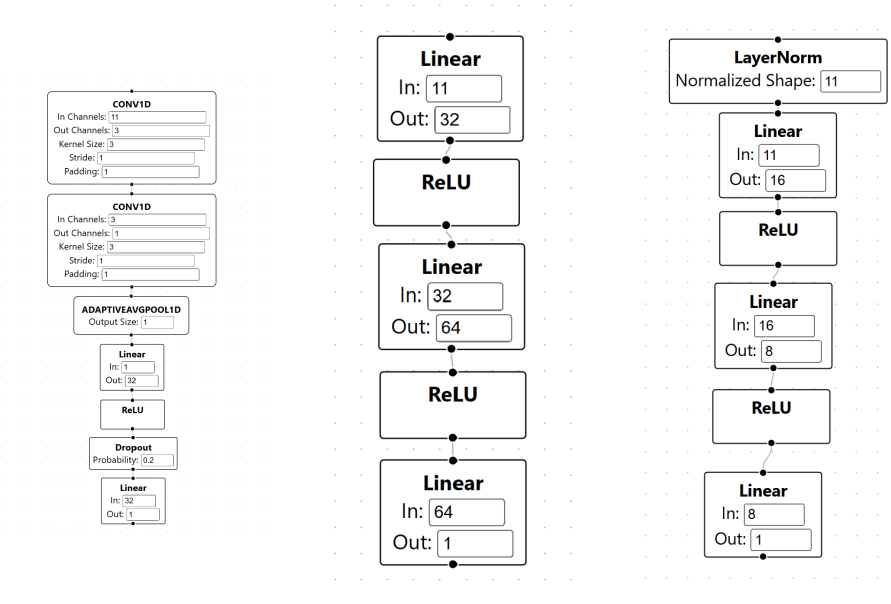

Figure 1: Model architectures

Conclusion: The three architectures represent different levels of model complexity — from a convolutional approach, through a standard MLP, to a normalized compact model. This allows analysis of the trade-off between capacity, stability, and generalization.

1.2 Dataset

The data used consists of real historical parameters for the years 2020 - 2023. Data from 2020 - 2022 were used to train the networks with different numbers of epochs and learning rates. The most suitable model is selected and then tested on the data from 2023. The parameters for each sample are:

- US/EU OLP - The base interest rate of the United States, set by the Federal Reserve / of the European Central Bank.
- US/EU Unemployment - The unemployment rate in the United States / European Union, expressed as a percentage.
- US/EU PMI - Purchasing Managers' Index for the United States / European Union. Values above 50 indicate economic expansion, while values below 50 indicate contraction in manufacturing activity.
- US/EU TradeBalance - The trade balance of the United States / European Union, measured as the difference between exports and imports.

- US/EU Inflation - The annual inflation rate in the United States / European Union.
- Oil Brent - The price of Brent crude oil, measured in USD per barrel.

Conclusion: The macroeconomic indicators cover key aspects of monetary policy, labor markets, economic activity, and foreign trade in the United States and the European Union, in order to model their impact on the exchange rate EUR-USD.

2 Training

We train each model on a data set from the first three years (2020–2022), using different numbers of epochs and learning rates. The neural networks are trained for 20, 100, 500 and 1000 epochs with learning rates of 0.01, 0.001, and 0.0001. The tables below illustrate the loss for each experiment.

2.1 Model 1

Epochs \ LR	0.01	0.001	0.0001
20	0.14474	0.16607	1.62229
100	1721824605896700.0	0.07916	1.02302
500	219.72842	0.04782	0.17621
1000	0.00442	0.02960	0.12499

Table 1: Model 1 error at different epochs and learning rates

Note: For a learning rate of 0.001 at 100, 500, and 1000 epochs, mode collapse occurs.

2.2 Model 2

Epochs \ LR	0.01	0.001	0.0001
20	1.06324E+23	0.08078	0.23478
100	5.16640E+31	0.01270	0.02378
500	0.00442	0.07652	0.00457
1000	1.05592E+28	0.00279	0.00984

Table 2: Model 2 error at different epochs and learning rate

Note: Mode collapse occurs at a learning rate of 0.01 for 20, 100, 500 and 1000 epochs, and at a learning rate of 0.001 for 500 epochs.

Epochs \ LR	0.01	0.001	0.0001
20	0.05247	0.75053	0.99117
100	0.00191	0.73679	1.07769
500	0.00436	0.02204	1.02995
1000	0.00268	0.00705	0.98525

Table 3: Model 3 error at different epochs and learning rate

2.3 Model 3

Note: Mode collapse occurs at a learning rate of 0.01 for 20, 100, 500, and 1000 epochs.

3 Validation

The validation is carried out in stages. First, unstable models—where mode collapse occurs—are removed. The second stage involves removing networks for which the error decreases as the number of epochs increases, indicating overfitting. From the remaining models, the one with the lowest error achieved is selected for each of 20, 100, 500, and 1000 epochs. The selected networks then proceed to the testing phase.

3.1 Stage 1

The following models were excluded:

- Model 1: learning rate, epochs = 0.001, 100, 0.001, 500, 0.01, 1000,
- Model 2: learning rate, epochs = 0.01, 20, 0.01, 100, 0.01, 500, 0.01, 1000, 0.001, 500
- Model 3: learning rate, epochs = 0.01, 20, 0.01, 100, 0.01, 500, 0.01, 1000

3.2 Stage 2

After the second stage, the following are eliminated: Model, learning rate = Model 1, 0.01, Model 2, 0.001, Model 3, 0.0001.

3.3 Stage 3

After comparing the total loss, the models that proceed to testing are: Model, learning rate, epochs = Model 1, 0.001, 20, Model 1, 0.0001, 100, Model 2, 0.0001, 500, Model 3, 0.001, 1000.

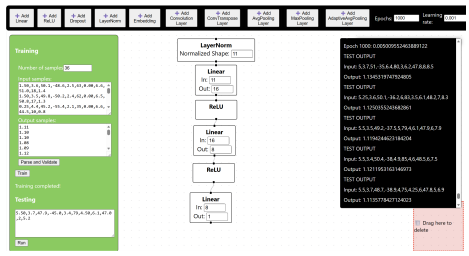

Figure 2: Final Model

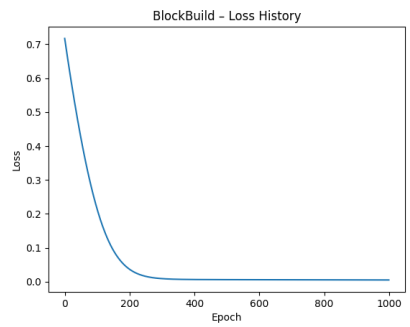

Figure 3: Loss function plot

4 Testing

In the final stage, we test the examples in the testing dataset and calculate their error using Mean Squared Error (MSE).

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where y_i is the real value, $\hat{y}_i$ is the predicted value, and n is the number of examples $n = 36$. Structure: Model, Learning rate, Epochs
Error for Model 1, 0.001, 20: 0.0774
Error for Model 1, 0.0001, 100: 0.0363
Error for Model 2, 0.0001, 500: 0.0705
Error for Model 3, 0.001, 1000: 0.0011
Conclusion: The most suitable model for predicting the EUR–USD exchange rate is Model 3 with a learning rate of 0.001 and 1000 epochs Fig.2. Loss function plot is presented in Fig. 3

5 Conclusion

Through this study, the most suitable network for the problem — a multilayer perceptron with data normalization — was selected, after examining three architectures across 12 possible configurations (learning rate and epochs). The datasets come from the recent years 2020–2023, demonstrating the model's accuracy on contemporary data and confirming its applicability. The parameters used are significant macroeconomic factors, which are generally accepted to have the greatest impact on the respective exchange rate. By determining the optimal structure, the study was successfully completed.

In the future, the range of considered architectures, configurations, macroeconomic factors, and training/testing examples will be expanded, which is expected to further improve the model's accuracy.